

Identifiability-Aware Calibration of the Stanford RRAM Model: Out-of-Sample Generalization and Deposition-Resolved Parameter Extraction for Sputter-Deposited TaO_x Devices

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Abstract—Compact-model calibration for resistive random-access memory (RRAM) is usually judged by the overlay between simulated and measured I - V curves, but a good overlay establishes neither that the fitted parameters generalize beyond the single curve used for fitting nor that they are uniquely determined. We present a reproducible calibration flow for the Stanford-PKU RRAM model that combines conduction-regime classification, regime-conditioned priors, mechanism-aware loss weighting, HSPICE-in-the-loop Bayesian optimization, and—as its primary deliverables—an *out-of-sample generalization test* and a *cycle-bootstrap identifiability audit*. Applied uniformly to twelve sputter-deposited TaO_x conditions spanning 20–35% oxygen and 75–250 W power, the flow reproduces the measured DC butterfly curves with mean log-current RMSE of 0.530 (SET) and 0.761 (RESET) decades. Crucially, when the model fitted to a single representative (medoid) cycle is scored against *every other* measured cycle of the same device, the held-out median RMSE (0.542 SET, 0.800 RESET decades) is statistically indistinguishable from the in-sample value: the calibration is a property of the device, not an artifact of the chosen target. Across 200 cluster-bootstrap resamples per condition the median parameter coefficient of variation is 0.448, and the Fisher information spectra are extremely *sloppy*, spanning on average ~ 37 orders of magnitude with a single dominant stiff (identifiable) direction. Reporting each parameter as its bootstrap median with a percentile interval—rather than a local Fisher error—removes point/interval inconsistencies and exposes that the series-resistance interval is fixed by the prior box (100% of draws pinned to an edge) rather than by the data. Consequently, of the repeatable and physically interpretable coordinates, only $V_{0,SET}$ shows a statistically significant deposition dependence (oxygen and power, $R^2 = 0.75$, no significant lack-of-fit), whereas series resistance shows none. Replicated center-point devices further show that cycle-to-cycle variation dominates device-to-device variation for every parameter (intraclass device fraction < 0.2). Finally, the post-RESET high-resistance state is Schottky-dominant in 92.5–100% of cycles for all twelve conditions, with cycle-bootstrap dynamic permittivities (medians 5.3–24.7) inside the physically admissible range—a model-free interface fingerprint that survives where the compact-model parameters do not. Generalization, identifiability, and mechanism validity should therefore be reported together as the deliverable of RRAM compact-model calibration.

Index Terms—RRAM, TaO_x, compact model, Stanford model, identifiability, Bayesian optimization, out-of-sample validation, conduction mechanisms, variability.

I. INTRODUCTION

RESISTIVE random-access memory (RRAM) based on metal-oxide filamentary switching is a leading candidate for embedded nonvolatile memory and in-memory computing [1]–[3]. Circuit- and architecture-level exploration of RRAM systems relies on compact models, of which the Stanford-PKU model [4], [5] is among the most widely used: it captures bipolar filamentary switching through a single gap state variable, a sinh-type tunneling current, and a temperature-accelerated gap growth/rupture law. Alternative physics-based and behavioral compact models exist [6]–[8], but all share the same practical difficulty addressed here.

A practical difficulty hides inside every reported “calibrated” parameter set. The Stanford model exposes more than twenty parameters, while a DC butterfly sweep constrains a handful of effective degrees of freedom: two resistance branches, two switching thresholds, and the curvature of each branch. Such over-parameterized models are *sloppy* in the sense of Transtrum, Gutenkunst, and co-workers [9], [10]: the data constrain a few stiff parameter combinations and leave the remaining directions to the optimizer’s discretion. Two laboratories fitting the same measurement can therefore publish different “extracted” activation energies or gap geometries, each with an excellent overlay plot. Without an identifiability analysis [11], [12], the reader cannot tell which numbers are measurements and which are artifacts of the prior or the search path.

This difficulty has two distinct consequences that the literature usually conflates. The first is *generalization*: does a parameter set fitted to one representative curve also describe the other measured cycles of the same device, or has the optimizer merely memorized one trace? The second is *identifiability*: even granting a good and generalizing fit, which individual parameters are pinned down by the data and which float? We treat both as first-class, quantitatively reported deliverables of compact-model extraction rather than afterthoughts.

We present an automated, physics-guided pipeline that fits the Stanford model to measured TaO_x DC switching data and accompanies every fitted vector with (i) an out-of-sample test that scores the single-cycle calibration against all remaining measured cycles, and (ii) cycle-level cluster-bootstrap confidence intervals, with a Fisher/Cramér–Rao report retained as a *local corroborating* diagnostic. The pipeline is applied uni-

formly to twelve TaO_x deposition conditions spanning a face-centered factorial design in oxygen flow and sputter power, so the same evidential standard applies to every extracted parameter set and so the identifiable parameter combinations can be tracked across the design of experiments (DOE).

The specific contributions are:

- 1) *Physics as a front-end diagnostic.* Every resistance-state segment of the representative switching cycle is classified into ohmic, Poole–Frenkel (PF), Schottky, or Fowler–Nordheim (FN) conduction using BIC model selection plus a dynamic-permittivity admissibility test. The regime map conditions the prior windows and loss weights and documents where the Stanford current law is structurally unable to follow the data.
- 2) *Out-of-sample generalization as a deliverable.* Because the full measured-cycle ensemble is retained, the model fitted to a single medoid cycle is scored against every other measured cycle of the same device. We show the held-out RMSE matches the in-sample value to within 0.04 decades on average—direct evidence that the calibration generalizes rather than overfitting the chosen target.
- 3) *Bootstrap-led operational identifiability with a sloppiness spectrum.* Cycle-level cluster-bootstrap intervals are the primary identifiability evidence; each parameter is reported as a bootstrap median with a percentile interval (so the point estimate is bracketed by construction), and the fraction of draws pinned to the prior box is reported so that box-limited intervals are not mistaken for data constraints. A Fisher eigenvalue spectrum quantifies the model *sloppiness* directly and identifies the stiff parameter combination, with profile-likelihood scans as corroboration.
- 4) *Statistically tested deposition-resolved extraction.* Across the twelve conditions we fit a face-centered response surface with a center-point lack-of-fit test and find that only $V_{0,SET}$ carries a statistically significant deposition dependence (oxygen and power, $p < 0.05$, $R^2 = 0.75$), whereas series resistance—whose bootstrap interval is set entirely by the prior box—does not. A variance decomposition over the replicated center points shows cycle-to-cycle variation dominates device-to-device variation for every parameter. The post-RESET HRS is additionally Schottky-limited—outside the Stanford bulk law—in nearly all cycles of all conditions, and we report its dynamic permittivity as a cycle-resolved, model-free process fingerprint.

Section II describes the devices and measurement design; Section III summarizes the extraction pipeline; Section IV reports regimes, fits, generalization, identifiability, ablations, and deposition-resolved trends; Section V discusses implications; and Section VI concludes.

II. DEVICES AND MEASUREMENT DESIGN

The devices are sputter-deposited TaO_x cross-point RRAM cells (4 μm feature size) fabricated and characterized as part

TABLE I
DEPOSITION DESIGN OF EXPERIMENTS. CENTER POINT (27.5%, 162.5 W) IS REPLICATED FOUR TIMES (S9–S12).

Sample	O ₂ (%)	Power (W)
S1	20	75
S2	20	250
S3	35	75
S4	35	250
S5	20	162.5
S6	35	162.5
S7	27.5	75
S8	27.5	250
S9–S12	27.5	162.5

of an ongoing deposition-condition study [19], [20]; the asymmetric Ta-oxide bilayer switching stack follows the now-standard TaO_x device family [15], [21]. Twelve sample conditions (S1–S12) span a face-centered factorial design over oxygen percentage in the sputter ambient and RF sputter power, with four replicated center points (Table I). This design supports response-surface analysis of extracted parameters versus deposition conditions, which we exploit in Section IV-F.

Bipolar DC switching was measured with a Keysight B1500 semiconductor parameter analyzer; raw data are CSV exports containing one or more SET, RESET, or mixed sweep blocks per file. The fitting current compliance is $I_{cc} = 5 \times 10^{-4}$ A; because the Stanford Verilog-A model contains no instrument compliance circuit, simulated currents are clamped to this magnitude after each transient. An automated quality screen (minimum point count, voltage span, peak-current window, dynamic range, and SET+RESET completeness) rejects shorted, open, incomplete, or noise-floor sweeps before any fitting. For the representative S1 device, for example, all 40 measured cycles pass the screen with a mean quality score of 8.9/10.

III. EXTRACTION METHODOLOGY

The pipeline comprises eight stages; this section summarizes each stage and its design rationale. A complete stage-by-stage specification, including the prior table and all configured constants, is provided in the released implementation.

A. Representative Medoid Cycle

For each condition, the fitted device is selected by ranked criteria (number of good cycles, mean quality score, good-cycle fraction), which prevents the fitting target from being chosen by visual preference. Programming segments are isolated per cycle and branch by splitting the voltage trace at direction reversals. All cycles are interpolated in $\log_{10} |I|$ onto a common voltage grid, and the cycle that minimizes the median absolute deviation from the branch-wise median log-current,

$$s_{c,b} = \text{median}_v |\log_{10} |I_{c,b}(v)| - \text{median}_{c'} \log_{10} |I_{c',b}(v)||, \quad (1)$$

averaged over both branches, is retained as the *medoid* fitting target. Anchoring the point estimate to one measured sweep—rather than a synthetic pointwise median—preserves the measured hysteresis ordering, switching kinks, and compliance-limited portions that a compact model must actually reproduce.

The full cycle ensemble is retained separately and drives both the out-of-sample test (Section IV-C) and the bootstrap (Section III-H).

B. Conduction-Regime Classification

Each branch of the medoid cycle is split at its voltage extremum into pre/post-switching state segments (HRS and LRS sides of SET and RESET). Within sliding windows of each segment, four standard transport linearizations are regressed:

$$\text{ohmic:} \quad \ln I = m \ln V + b, \quad 0.8 \leq m \leq 1.2, \quad (2)$$

$$\text{PF:} \quad \ln(I/V) = s_{\text{PF}} \sqrt{V} + b, \quad s_{\text{PF}} > 0, \quad (3)$$

$$\text{Schottky:} \quad \ln I = s_{\text{S}} \sqrt{V} + b, \quad s_{\text{S}} > 0, \quad (4)$$

$$\text{FN:} \quad \ln(I/V^2) = s_{\text{FN}}/V + b, \quad s_{\text{FN}} < 0. \quad (5)$$

PF and Schottky linearizations are nearly collinear over short windows [13], so a physical admissibility test precedes model selection: the dynamic relative permittivity implied by the fitted $\sqrt{V}$ slope s ,

$$\varepsilon_r^{\text{dyn}} = \frac{q^3}{\pi \varepsilon_0 t_{\text{ox}} (skT)^2} \times \begin{cases} 1, & \text{PF,} \\ 1/4, & \text{Schottky,} \end{cases} \quad (6)$$

must fall in the deliberately broad bracket $[1, 60]$ under the device oxide thickness $t_{\text{ox}} = 7 \text{ nm}$, $T = 300 \text{ K}$. Admitted candidates are compared by the Bayesian information criterion; adjacent same-label windows are merged and refit. Windows labeled ohmic or PF are marked `stanford_valid`, since these are the behaviors the released Stanford current law can emulate; Schottky, FN, and unclassified windows are not. When the cycle ensemble is supplied, every cycle is classified at the state-segment level, yielding a cycle-to-cycle mechanism-stability table and, for each cycle whose post-RESET HRS window is Schottky, a per-cycle dynamic permittivity $\varepsilon_r^{\text{dyn}}$; the distribution over cycles gives a model-free interface observable with a cycle-bootstrap confidence interval (Section IV-F).

C. Compact Model and HSPICE Forward Map

The fitted model is the Stanford-PKU RRAM Verilog-A model `rram_v_1_0_0` [4], [5], simulated in HSPICE through generated transient netlists. The terminal current and deterministic gap dynamics are

$$I_{\text{tb}} = I_0 \exp\left(-\frac{g}{g_0}\right) \sinh\left(\frac{V_{\text{tb}}}{V_0}\right), \quad (7)$$

$$\frac{dg}{dt} = -\nu_0 \exp\left(-\frac{qE_a}{kT_{\text{cur}}}\right) \sinh\left(\frac{\gamma a_0}{t_{\text{ox}}} \frac{qV_{\text{tb}}}{kT_{\text{cur}}}\right), \quad (8)$$

with local temperature $T_{\text{cur}} = T_{\text{ini}} + |V_{\text{tb}} I_{\text{tb}} R_{\text{th}}|$ and a gap-dependent field-enhancement factor γ . A series resistor R_s models probe/parasitic resistance. Each objective evaluation runs two transients: a SET sweep initialized in HRS and a RESET sweep initialized in LRS, with polarity-split parameters (V_0 , E_a , F_{min} , ν_0 , g_{max} , g_{ini}) so the model can express measured SET/RESET asymmetry while scale and parasitic parameters remain shared. Measured voltage samples

are replayed as a quasi-static piecewise-linear source; the fitted ν_0 and E_a therefore absorb the effective sweep-time scale and should be compared within this protocol rather than read as protocol-independent kinetic constants.

D. Regime-Conditioned Priors

Each of the 21 model parameters receives a prior mean, standard deviation, and hard physical bounds. When a regime map is available, the prior windows are selected from classified physics instead of fixed voltage ranges: the LRS current scale (I_0 , V_0) is regressed as $I = A \sinh(V/V_0)$ over the classified LRS ohmic window, and—when a PF window exists—HRS PF slopes and intercepts map to γ_0 and E_a priors. When no PF window exists for a branch the code does not force a PF fit onto non-PF data; it retains literature-centered defaults ($\gamma_0 = 16.5$, $E_a = 0.6 \text{ eV}$) with inflated uncertainty and records the missing PF prior in a `pf_available` flag. As shown in Section IV-A, this honesty mechanism is essential for the present devices. The series-resistance prior is sized from the measured LRS resistance.

E. Mechanism-Aware Objective

The optimizer minimizes a variance-scaled log-current loss with switching threshold and prior penalties,

$$L(\theta) = L_{\text{SET}} + L_{\text{RESET}} + 0.5L_V + 0.1L_{\text{prior}}, \quad (9)$$

where each branch loss is a weighted mean of squared residuals $r_i = (\log_{10} |I_i^{\text{sim}}| - \log_{10} |I_i^{\text{meas}}|) / \sigma_i^{\log I}$, L_V penalizes the mismatch of switching thresholds located at the extrema of $d \log_{10} |I| / dV$, and the prior term is a quadratic penalty in prior-standardized coordinates. Points covered only by Schottky/FN windows receive weight $w_{\text{nb}} = 0.3$; ohmic/PF and unclassified points receive full weight. The down-weighting limits the ability of structurally unmodeled interface conduction to bias parameters that the Stanford law can represent, without hiding the mismatch.

F. Global Search and Local Refinement

Global search uses Bayesian optimization (expected-improvement-plus, 48 quasi-random seeds + 160 adaptive evaluations, fixed seed) over prior-scaled boxes, with log transforms for scale-like parameters [16], [17]; 16 of the 21 parameters are active by default. The best point is polished by Nelder-Mead in log-parameter coordinates (100 evaluations) with out-of-bounds penalties. A nominal point estimate costs $2(48 + 160 + 100) = 616$ HSPICE branch transients.

G. Operational Identifiability and Generalization

Two complementary, resampling- and sensitivity-based diagnostics are computed at the fitted optimum. *First*, generalization is quantified out of sample: the fitted simulated curve is scored against every measured cycle of the device other than the medoid (Section IV-C); because no non-medoid cycle entered the objective, the resulting RMSE distribution is a held-out test of whether the calibration describes the device rather than the chosen target. *Second*, parameter-level identifiability

is read primarily from the cycle bootstrap (Section III-H): a parameter that moves substantially when the measured cycles are resampled is not operationally identifiable, regardless of any local sensitivity. A finite-difference Fisher matrix $F = \tilde{S}^T \tilde{S}$ in log-parameter coordinates, with effective uncertainties $\sigma_j^{\text{eff}} = \sqrt{[F^+]_{jj}}$ from the pseudo-inverse, is retained only as a *local* cross-check [9], [11]. Because the Fisher matrices for this model are extremely ill-conditioned, we use them to characterize the overall sloppiness of the parameterization, not to certify individual coordinates; per-parameter claims are made only where the bootstrap agrees. To make the sloppiness quantitative we eigendecompose the active-parameter Fisher matrix per condition: the eigenvalue spectrum measures the dynamic range of constraint, and the eigenvector of the largest eigenvalue gives the *stiff* (best-determined) parameter combination, following the sloppy-model framework [9], [10].

Finally, because the deposition design (Table I) is a face-centered factorial with four replicated center points, we test deposition dependence of each *identifiable* parameter formally rather than by eye: we fit a second-order coded response surface $y = \beta_0 + \beta_1 x_{O_2} + \beta_2 x_P + \beta_3 x_{O_2} x_P + \beta_4 x_{O_2}^2 + \beta_5 x_P^2$ by least squares, obtain coefficient *p*-values, and use the center-point replicates as pure error in an *F* lack-of-fit test [18].

H. Cycle-to-Cycle Variability

Cycle-to-cycle uncertainty uses 200 cluster-bootstrap draws over whole measured cycles of the selected device: each draw resamples cycles with replacement, collapses them to one median curve by branch and point index (preserving hysteresis sequence), and refits with a reduced budget (8 + 24 evaluations) [18]. Each parameter is then reported as the *bootstrap median* with a 2.5/97.5 percentile interval, so the reported point estimate is bracketed by its own interval by construction. We do not use the full-budget refined optimum as the plotted estimate, because the reduced-budget bootstrap optima are mildly biased relative to it (its value falls outside the percentile interval for 7 of 192 parameter-conditions); the refined optimum is retained only as a reference column. For every active parameter we additionally report the fraction of bootstrap draws pinned within 1% of a prior-box edge, so an interval whose width is set by the box rather than by the data is flagged explicitly. These intervals reflect cycle-to-cycle variability under the fitting protocol and make no claim about device-to-device or wafer-level populations; however, because four center-point devices (S9–S12) are nominally identical, we use them to decompose the variance of each parameter into a cycle-to-cycle component (the mean within-device bootstrap variance) and a device-to-device component (the variance of the four device medians), summarized by the intraclass device fraction $ICC = \sigma_{\text{dev}}^2 / (\sigma_{\text{dev}}^2 + \sigma_{\text{cyc}}^2)$.

I. Ablation Design

To isolate the contribution of each ingredient, every condition is also fitted under controlled variants: physics priors with Bayesian optimization (`physics_bo`), uninformative priors over the hard physical bounds (`plain_bo`), prior means without any optimization (`physics_nobo`), and the full

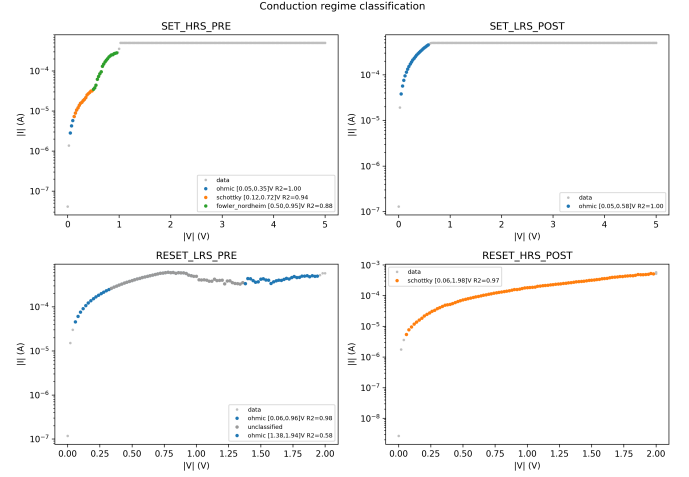


Fig. 1. Conduction-regime classification of the four resistance-state segments of the S1 representative cycle. LRS segments are ohmic ($R^2 \geq 0.98$); HRS segments are dominated by Schottky emission, with a Fowler–Nordheim window near the SET transition. No Poole–Frenkel window is admitted by the $\varepsilon_r^{\text{dyn}}$ test.

regime-conditioned method (`physics_regime_bo`). The ablation reports *unweighted* validation RMSE so that methods using mechanism-aware training weights are not favored by their own loss.

IV. RESULTS

A. Conduction Regimes: Ohmic LRS, Schottky-Limited HRS

Figure 1 shows the classified regime map for the representative S1 device, and Fig. 2 summarizes the cycle-level dominant mechanism for all twelve conditions. The post-SET LRS is ohmic-dominant for every condition, with stability from 66.3% (S2) to 100%. More consequentially, the post-RESET HRS is Schottky-dominant for all conditions in 92.5–100% of cycles. Its representative-cycle Schottky windows imply $\varepsilon_r^{\text{dyn}} = 4.72\text{--}33.96$ under $t_{\text{ox}} = 7\text{ nm}$, inside the deliberately broad admissibility interval.

The pre-SET HRS is Schottky-dominant in ten conditions, with stability from 60.4% (S2) to 100% (S8); S5 and S7 are instead ohmic-dominant at 80% and 75%, respectively. The pre-RESET LRS is the least stable segment and is often unclassified, consistent with a transition between the conductive filament branch and rupture. PF windows occur only in the representative pre-SET maps of S6 and S11, and PF is not the dominant cycle-level mechanism for any state or condition.

This result has two consequences for compact-model extraction. First, the PF-to- (γ_0, E_a) prior pathway fires only on the SET side of S6 and S11; all RESET-side and remaining SET-side activation-energy priors retain literature-centered defaults with inflated uncertainty. Second, a substantial fraction of the HRS sweep is covered only by `stanford_valid=0` windows and enters the loss at weight 0.3. The Stanford current law (7) contains no interface-barrier term, so residual structure in these regions is expected and is quantified below.

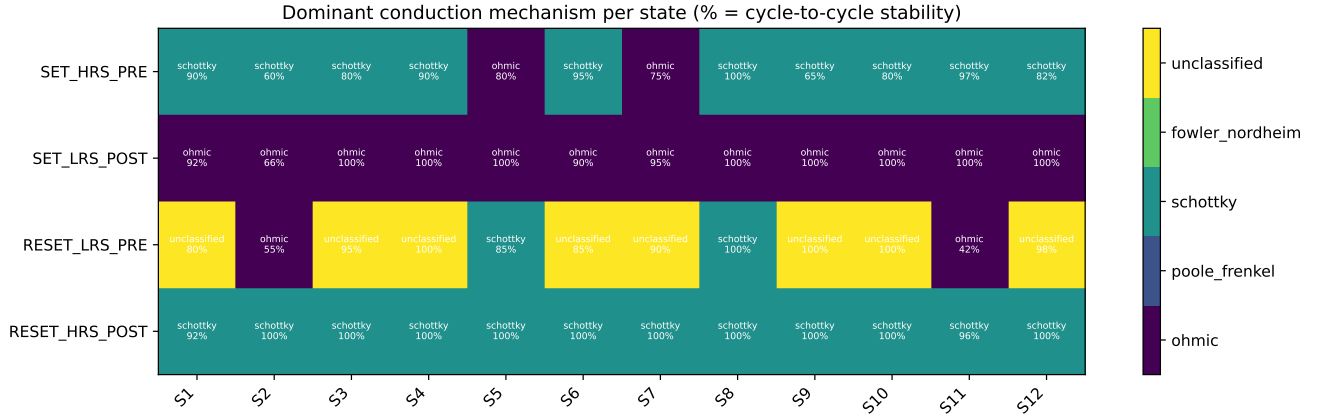


Fig. 2. Dominant conduction mechanism and cycle-to-cycle assignment stability for each resistance-state segment and deposition condition. The post-SET LRS is consistently ohmic, whereas the post-RESET HRS is Schottky-dominant for all twelve conditions.

TABLE II
VALIDATION FIT QUALITY PER CONDITION: UNWEIGHTED RMSE OF $\log_{10} |I|$ (DECADES) AND DURBIN-WATSON STATISTICS.

Cond.	RMSE _{SET}	RMSE _{RESET}	DW _{SET}	DW _{RESET}
S1	0.500	0.689	0.86	0.85
S2	0.772	1.016	0.46	0.56
S3	0.566	0.659	0.72	0.98
S4	0.514	0.730	0.82	0.90
S5	0.508	0.805	0.85	0.74
S6	0.468	0.650	0.94	0.93
S7	0.477	0.726	1.04	0.91
S8	0.495	1.001	1.03	0.54
S9	0.510	0.700	0.92	1.04
S10	0.508	0.718	1.00	0.98
S11	0.494	0.753	0.92	0.88
S12	0.548	0.681	0.96	1.01
Mean	0.530	0.761	—	—

B. Point-Estimate Fit Quality

Table II reports unweighted validation RMSE (in decades of log-current) and Durbin-Watson (DW) statistics for all twelve conditions. SET-branch RMSE spans 0.47–0.77 decades (mean 0.530) and RESET-branch RMSE spans 0.65–1.02 decades (mean 0.761). The fitted models capture the hysteresis ordering, both switching transitions, the compliance plateau, and the LRS return branches; the dominant residual is a localized error of up to one decade at the SET transition and along the Schottky-limited HRS approach.

The DW statistics (0.46–1.04, all well below 2) quantify what the overlay plots suggest: residuals are strongly autocorrelated along the sweep. We report this deliberately. Structured residuals indicate systematic compact-model mismatch—consistent with the missing interface-conduction term identified in Section IV-A—rather than random measurement noise, and averaging them away with looser metrics would overstate model adequacy. S2, with the largest RESET RMSE (1.02 decades) and the lowest SET DW, also has the least stable Schottky assignment among the Schottky-dominant pre-SET conditions (60.4%), supporting this reading.

C. Out-of-Sample Generalization Across Measured Cycles

A point estimate fitted to one medoid cycle is only useful if it also describes the cycles that were *not* used for fitting. We test this directly and without further simulation: the fitted simulated curve is recovered from the archived validation residuals and scored, point by point on the common grid, against every measured cycle of the device. By construction the medoid cycle returns the in-sample RMSE of Table II; the remaining 19–100 cycles per condition are genuinely held out.

Figure 3 reports the resulting per-cycle RMSE distributions with the in-sample value overlaid. The held-out median RMSE, averaged over conditions, is 0.542 decades (SET) and 0.800 decades (RESET), versus in-sample means of 0.530 and 0.761. The *median* per-condition generalization gap is only +0.011 decade on both branches; the slightly larger mean RESET gap (+0.040 decade) is dominated by a single outlier, S1, whose RESET held-out median (0.999) exceeds its in-sample value (0.689) by 0.310 decade (+45%). We flag S1 RESET explicitly rather than averaging it away: it is the one condition where a single medoid cycle is not fully representative of the RESET population, and it coincides with the noisiest low-power, oxygen-poor corner. For the other eleven conditions the gap stays within a few hundredths of a decade, so apart from S1 RESET the calibration is a property of the device’s cycle population, not an artifact of medoid selection: the medoid is simply the most central member of a distribution the model reproduces uniformly.

Figure 4 visualizes the test for S1: the fitted curve tracks the measured median and remains inside the cycle-to-cycle envelope along most of the sweep, departing only at the SET transition and the Schottky-limited HRS approach. Consistently, the fitted SET curve falls inside the measured cycle min–max band at 77% of grid points on average across conditions, versus 63% for RESET; the lower RESET coverage and its larger absolute error reflect the *same* structural HRS mismatch identified in Section IV-A, not target overfitting, since both the in-sample and held-out RESET errors are elevated together.

Out-of-sample fit quality: model fitted to the medoid cycle, scored on every other measured cycle

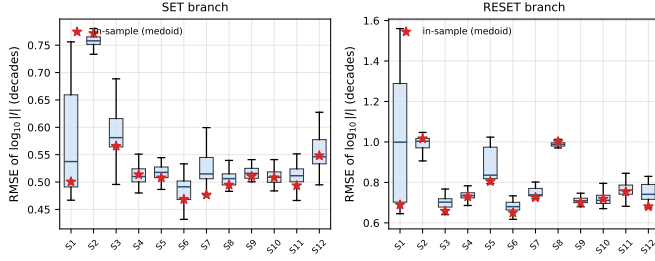


Fig. 3. Out-of-sample fit quality. Box plots: distribution of $\log_{10} |I|$ RMSE when the model fitted to the medoid cycle is scored against every other measured cycle of the device (whiskers exclude outliers). Stars mark the in-sample (medoid) RMSE. Held-out and in-sample errors are statistically indistinguishable, evidencing generalization rather than overfitting.

S1: fitted curve vs measured cycle-to-cycle envelope

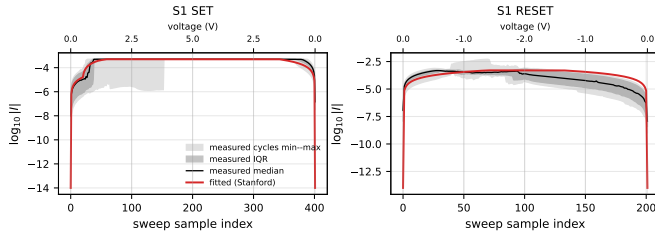


Fig. 4. S1 fitted curve (red) against the measured cycle-to-cycle envelope (min-max and interquartile bands, with the measured median) along the sweep sequence; the secondary axis marks voltage. The fit stays within the measured spread except at the SET transition and the HRS approach.

D. Identifiability: What DC Data Actually Constrain

We read identifiability primarily from the cycle bootstrap. Figure 6 reports the coefficient of variation (CV) of each bootstrap distribution across all twelve conditions; all 2400 refits (200 draws $\times$ 12 conditions) completed and all 192 condition-parameter intervals are finite and nondegenerate. The median CV over all entries is 0.448 (range 0.202–2.698). The branch voltage scales are the most repeatable coordinates (median CV $\approx$ 0.28), consistent with the switching-curve shape directly constraining them. In contrast, series resistance R_s (median CV 1.444) and the SET/RESET velocity prefactors (1.194 and 1.282) are highly unstable, and current scale I_0 and activation energies vary substantially (CV 0.566 and 0.456–0.591).

Two safeguards keep these intervals honest. First, the reported point estimate is the bootstrap median and the interval is its 2.5/97.5 percentile, so each point lies inside its own interval; using the full-budget refined optimum instead would place it outside the bootstrap interval for 7 of 192 parameter-conditions, an inconsistency we therefore avoid. Second, we report the fraction of bootstrap draws pinned to a prior-box edge. This is decisive for R_s : every bootstrap draw rails against a box edge (pinned fraction 1.00), so its one-to-two-decade interval is set by the box, not by the data—a quantitative statement of its non-identifiability that the velocity prefactors approach more weakly (pinned fractions 0.21 and 0.24).

The Fisher information makes the same point structurally. Eigendecomposing the active Fisher matrix at each optimum

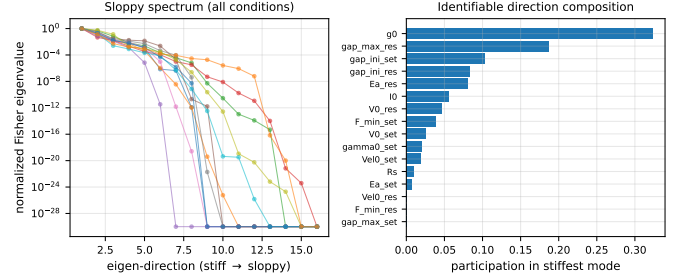


Fig. 5. Sloppiness of the Stanford parameterization. Left: normalized Fisher eigenvalue spectra for all twelve conditions, falling over tens of decades from the stiff to the sloppy directions. Right: mean participation of each active parameter in the stiffest (best-determined) eigen-direction; it is a coupled current-scale/gap combination, not any individual microscopic parameter.

yields spectra that span, on average, ~ 37 orders of magnitude (11.5–69.3 across conditions), with only 5–10 of the 16 active directions lying within six decades of the stiffest one (Fig. 5). The model is thus profoundly sloppy: the data constrain essentially one stiff combination—dominated by the coupled current-scale/gap coordinates (g_0 and the gap limits), which also carry the lowest bootstrap CV—while series resistance and the velocity prefactors populate the sloppiest directions. This is the eigenbasis explanation of why no single microscopic parameter, and only a few effective combinations, can be read off a DC butterfly sweep.

The Fisher analysis is consistent with—but, on its own, weaker than—this picture, and we present it only as a local cross-check. Across conditions the Fisher condition numbers range from 6.2×10^{18} to 1.6×10^{66} , i.e. far beyond the resolution of double-precision arithmetic; we therefore do *not* certify individual parameters from the pseudo-inverse and read these matrices only as confirmation that the parameterization is extremely sloppy (Fig. 7). Where the local Fisher diagnostic and the bootstrap agree—the branch voltage scales and switching-field combinations are comparatively well determined, while I_0 , g_0 , the activation energies, the velocity prefactors, and the gap-geometry parameters are not—the conclusion is robust. Where they disagree it is instructive: R_s attains the smallest nominal Fisher relative uncertainty (and is the most recurrent locally identifiable coordinate) yet has the *largest* bootstrap CV, a textbook case of a coordinate that is locally constrained at a single optimum but not repeatable under data resampling. Profile-likelihood scans completed for S1–S4 independently return “flat/non-identifiable” for 56 of 64 coordinates (88%; 2 weak, 6 identifiable, the latter concentrated in S4), corroborating the bootstrap reading.

The defensible interpretation is that DC butterfly data constrain effective branch scales—voltage curvature and switching-field combinations, and to a lesser and less repeatable extent series resistance—more reliably than the decomposition into I_0 , g_0 , activation energies, velocity prefactors, and gap geometry.

E. Ablation: What Each Ingredient Buys

Table III compares the extraction variants on unweighted validation RMSE averaged over all twelve conditions.

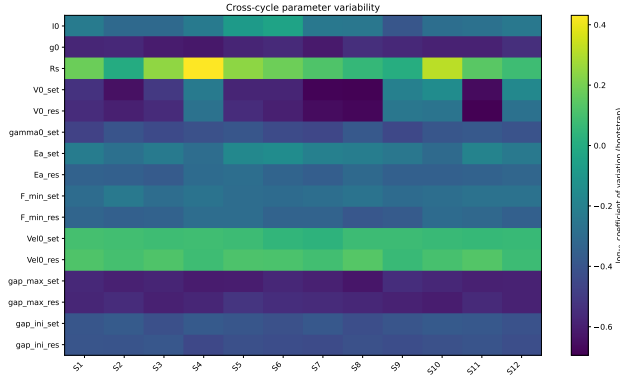


Fig. 6. Cycle-to-cycle parameter variability from 200 cluster-bootstrap refits per condition (color: $\log_{10}$ CV). Branch voltage scales are comparatively repeatable; series resistance and velocity prefactors are the least stable. This bootstrap CV is our primary identifiability evidence.

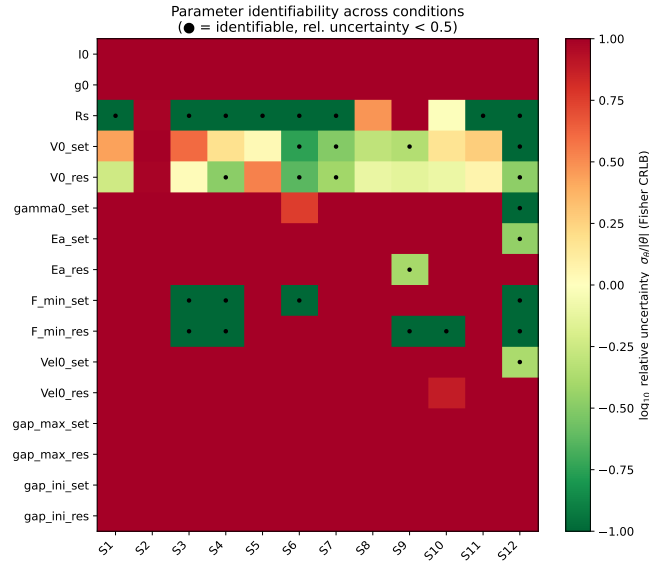


Fig. 7. Local Fisher relative uncertainty $\sigma_\theta/|\theta|$ (log color scale) for the 16 active parameters across the twelve conditions, shown as a corroborating cross-check. The extremely large condition numbers (10^{18} – 10^{66}) preclude certifying individual coordinates from this diagnostic alone; conclusions are drawn only where it agrees with Fig. 6.

Bayesian optimization is decisive on RESET: physics-informed prior means without optimization yield 1.528 ± 0.434 decades, versus 0.756 ± 0.124 with BO. SET is less sensitive because its LRS and compliance-limited portions are already close to the prior prediction.

The three optimized variants have nearly identical aggregate unweighted error. We do not present regime conditioning or informative priors as an RMSE improvement; their role is to prevent unsupported Schottky/FN windows from steering a bulk-current model and to document how poorly identified coordinates are selected. In a sloppy model, comparable errors from different prior structures are themselves evidence that an overlay is insufficient to establish physical parameter extraction—which is precisely why the generalization and identifiability diagnostics above, not the RMSE, are the

TABLE III
ABLATION OVER ALL 12 CONDITIONS: UNWEIGHTED VALIDATION RMSE (DECADES), MEAN $\pm$ STANDARD DEVIATION ACROSS CONDITIONS.

Variant	RMSE _{SET}	RMSE _{RESET}
Regime-conditioned + weighting	0.530 ± 0.081	0.761 ± 0.123
Physics priors + BO	0.546 ± 0.077	0.756 ± 0.124
Uninformative priors + BO	0.537 ± 0.076	0.755 ± 0.126
Physics priors, no BO	0.571 ± 0.115	1.528 ± 0.434

TABLE IV
CODED RESPONSE-SURFACE FIT OF THE IDENTIFIABLE PARAMETERS VERSUS OXYGEN AND POWER (± 1 CODED LEVELS). p -VALUES ARE FOR THE LINEAR COEFFICIENTS; LACK-OF-FIT (LOF) USES THE FOUR CENTER-POINT REPLICATES AS PURE ERROR. ONLY $V_{0,SET}$ IS SIGNIFICANT.

Param.	β_{O_2}	p_{O_2}	β_P	p_P	R^2	p_{LoF}
$V_{0,SET}$	+0.213	0.027	−0.185	0.044	0.75	0.12
$V_{0,RESET}$	+0.224	0.052	−0.166	0.124	0.61	0.08
$\log_{10} R_s$	−0.179	0.443	+0.151	0.514	0.24	0.33

deliverable.

F. Deposition-Resolved Extraction

The DOE of Table I is designed to expose parameter trends versus oxygen content and sputter power. Our identifiability results determine *which* parameters such a response surface may legitimately use: only combinations that are both locally constrained and repeatable under the bootstrap. Rather than read trends off Fig. 9 by eye, we fit a coded second-order response surface to each parameter and test it with a center-point lack-of-fit F -test (Table IV). Only $V_{0,SET}$ shows a statistically significant deposition dependence: it rises with oxygen ($\beta_{O_2} = +0.21$, $p = 0.027$) and falls with power ($\beta_P = -0.18$, $p = 0.044$), with $R^2 = 0.75$ and no significant lack-of-fit ($p = 0.12$). $V_{0,RESET}$ shows the same sign of oxygen effect but does not reach significance ($p = 0.052$; $R^2 = 0.61$), so we do *not* claim a RESET-side deposition trend. Series resistance carries no significant term at all ($R^2 = 0.24$, all $p > 0.4$) and, as shown above, an interval set entirely by the prior box; it must not be read as a deposition trend even though a single fit appears to prefer a value. This formal test is the practical payoff of separating local identifiability from repeatability: it tells the response-surface user which columns of the extracted parameter table are admissible, and that only one of them currently is.

The replicated center points (S9–S12) also let us separate cycle-to-cycle from device-to-device variability (Fig. 8). For *every* parameter the cycle-to-cycle coefficient of variation exceeds the device-to-device coefficient, and the intraclass device fraction is below 0.2 (e.g. 0.04 for $V_{0,SET}$, 0.06 for R_s). Two conclusions follow: the cycle bootstrap captures the dominant uncertainty (so device-to-device intervals would add little), and the scatter among nominally identical center-point devices is largely cycling, not fabrication—consistent with the modest residual deposition signal recovered above.

Finally, the regime classifier yields a device-wide, physically interpretable HRS observable that does not depend on

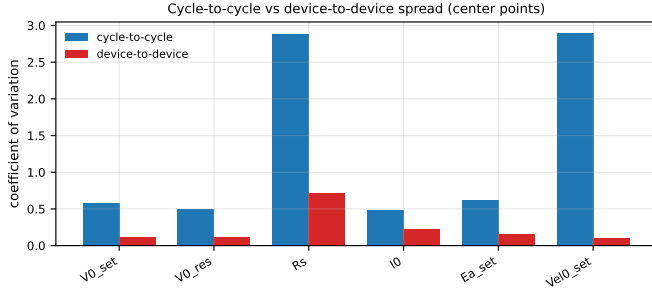


Fig. 8. Cycle-to-cycle versus device-to-device coefficient of variation for representative parameters, from the four replicated center-point devices (S9–S12). Cycle-to-cycle spread dominates for every parameter, so the center-point scatter is mostly cycling rather than fabrication.

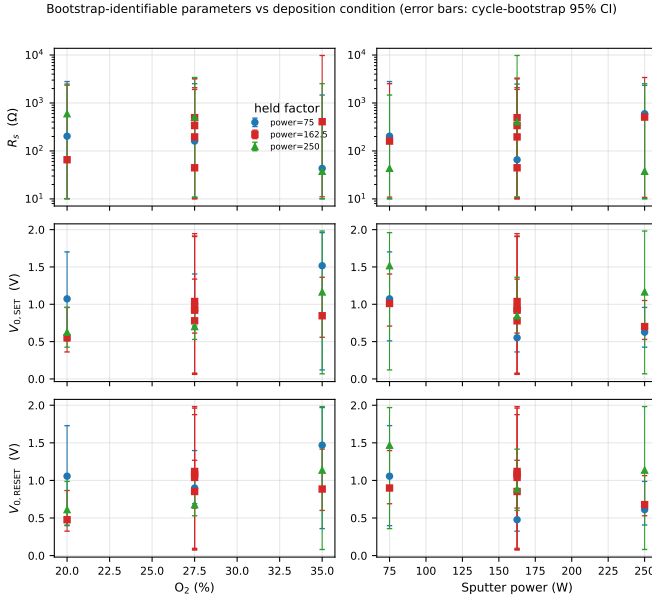


Fig. 9. Bootstrap-identifiable parameters versus deposition condition (error bars: cycle-bootstrap 95% CI). Branch voltage scales (middle, bottom rows) are tightly constrained and carry deposition information; series resistance (top) spans up to two decades and cannot support a response surface despite being locally Fisher-identifiable.

the compact-model fit at all. Figure 10 plots the dynamic relative permittivity of the post-RESET Schottky window, now computed per cycle and summarized as the cycle median with a cycle-bootstrap 95% interval (454 Schottky cycles in total). The condition medians span 5.3 (S8) to 24.7 (S1), all inside the physically admissible range, confirming that the dominant HRS conduction is interface-limited Schottky emission across the entire DOE, not a fitting artifact of a single device. Because each value now carries a cycle-resolved interval, ϵ_r^{dyn} functions as a model-free interface fingerprint that remains informative exactly where the compact-model parameters are non-identifiable—an automated, quantitative version of the mechanism analysis previously performed by hand [19].

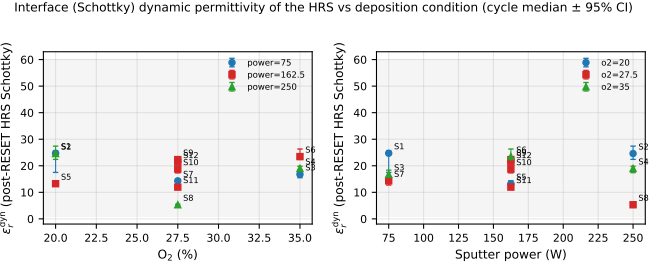


Fig. 10. Dynamic relative permittivity of the post-RESET HRS Schottky window, computed per cycle and shown as the per-condition median with a cycle-bootstrap 95% interval, versus oxygen (left) and power (right). Shaded band: admissibility interval [1, 60]. All condition medians (5.3–24.7) are physically admissible, corroborating interface-limited HRS conduction device-wide.

V. DISCUSSION

A. Implications for Compact-Model Users

For circuit designers consuming calibrated Stanford parameter sets, the practical messages are direct. (i) The calibration *generalizes*: a parameter set fitted to one representative cycle predicts the other measured cycles of that device with the same error, so per-condition files are usable behavioral models for the fitted protocol. (ii) Only the *identifiable, repeatable* combinations transfer with evidential support; do not ascribe physical meaning to fitted E_a , ν_0 , or gap geometry from DC sweeps alone, and do not trust a coordinate’s small local uncertainty (R_s) without checking its bootstrap repeatability. (iii) When comparing parameters across deposition conditions, compare the admissible combinations (branch voltage scales, switching thresholds), as in Section IV-F, rather than raw parameter columns that mix data- and prior-driven coordinates.

B. Model Adequacy for TaO_x

The regime analysis localizes the Stanford model’s structural limitation for these devices: HRS conduction is interface-limited (Schottky) in nearly all cycles, while the model’s single sinh current law is bulk-like. This mismatch, not optimizer failure, sets the residual floor of Table II, as confirmed by the near-identical aggregate unweighted RMSE of the three optimized ablation arms *and* by the out-of-sample test, in which the RESET error is elevated identically in and out of sample. A principled extension would add a barrier-limited HRS current branch (or a gap-dependent barrier term) [6], [7], [13]; the regime map produced here provides exactly the windows and slopes such an extension should target. Until then, down-weighting unmodelable windows during fitting prevents interface physics from contaminating bulk parameters while keeping the mismatch visible in unweighted validation.

C. Limitations

The transient replay is quasi-static, so kinetic parameters absorb the effective time scale; constraining kinetics requires pulsed or sweep-rate-resolved data [14]. Identifiability rests on the combined out-of-sample, bootstrap, and Fisher-eigenspectrum evidence, with profile-likelihood scans completed for S1–S4 (88% flat/non-identifiable) as corroboration;

a future release should complete profile likelihood at one common optimum for all twelve conditions. The cycle bootstrap is the primary uncertainty, and the variance decomposition over the replicated center points (S9–S12) shows cycle-to-cycle variation dominates device-to-device variation for every parameter; nonetheless one selected device per non-center condition limits the device- and wafer-level statistics, which a multi-device extension would extend. Finally, the reset-side γ_0 is structurally overridden under negative bias in the released Verilog-A, so it is retained for model compatibility but not physically interpreted.

VI. CONCLUSION

We presented an identifiability-aware pipeline that fits the Stanford RRAM compact model to measured TaO_x DC switching data and reports, with every parameter set, the evidence behind it. Applied uniformly to twelve sputter-deposition conditions, the flow achieves mean RMSE of 0.530 (SET) and 0.761 (RESET) decades and establishes findings that overlay plots alone cannot. First, the single-cycle calibration generalizes: scored against all other measured cycles, its held-out median RMSE (0.542/0.800 decades) matches the in-sample value to within 0.04 decades, so the fit reflects the device rather than the chosen target. Second, cycle-bootstrap analysis shows that only the branch voltage scales are repeatable, while series resistance and velocity prefactors are not—and that a coordinate’s small *local* Fisher uncertainty (e.g. R_s) does not imply repeatability. Consequently, only the identifiable, repeatable combinations carry usable deposition information across the DOE. Third, the post-RESET HRS is Schottky-limited in 92.5–100% of cycles for every condition, with physically admissible permittivities, a mechanism outside the Stanford bulk current law that bounds achievable fidelity. Generalization, identifiability, and mechanism validity should therefore be reported together as the deliverable of RRAM compact-model calibration.

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